Homework 8

Monday, April 6, 2020 4:19 PM

(1) FT - VIII

1. $CO + * \rightleftharpoons CO^*$
2. $H_2 + 2* \rightleftharpoons 2H^*$
3. $CO^* + 2H \rightleftharpoons HCOH^* + 2*$
4. $HCOH^* + * \rightleftharpoons CH^* + OH^*$
5. $CH^* + H^* \rightleftharpoons CH_2^* + *$
6. $OH^* + H^* \rightleftharpoons H_2O + 2*$

$r_1 = -k_1 (C_{CO} - C_{CO^*})$

$r_2 = -k_2 (C_{H_2} - C_{H^*})$

$r_3 = -k_3 (C_{CO^*} \cdot [C_{H^*}]^2 - C_{HCOH^*})$

$r_4 = -k_4 (C_{HCOH^*} - C_{CH^*} \cdot C_{OH^*})$

$r_5 = -k_5 (C_{CH^*} \cdot C_{H^*} - C_{CH_2^*})$

$r_6 = -k_6 (C_{OH^*} \cdot C_{H^*} - C_{H_2O})^2$

$$\frac{r_1}{k_1} = \frac{r_2}{k_2} = \frac{r_3}{k_3} = \frac{r_4}{k_4} = \frac{r_5}{k_5} = \frac{r_6}{k_6} = 0 \rightarrow$$

$C_{CO^*} = C_{CO_2}$ — (1)

$C_{H_2} = C_{H^*}$ — (2)

$C_{HCOH^*} = C_{CH^*} \cdot C_{OH^*}$ — (4)

$C_{CH^*} \cdot H^* = C_{CH_2^*}$ — (5)

$C_{OH^*} C_{H^*} = C_{H_2O}$ — (6)

$C_{OH^*} = \frac{C_{H_2O}}{C_{H^*}}$ (from 6)

$$r_3 = -k_3 (C_{CO^*} [C_{H^*}]^2 - C_{HCOH^*})$$

$$r_3 = -k_3 (C_O C_{H_2} - C_{CH^*} . C_{OH^*})$$

$$r = r_3 = -k_3 \left(C_O C_{H_2} - \frac{C_{CH^*} C_{H_2O}}{C_{H^*}} \right)$$

$$[CHCOH^*] = [CH^*][OH^*]$$

$$\boxed{r_3 = -k_3 \left([CO][H_2] - \frac{[CH^*][H_2O]}{[H^*]} \right)}$$

overall rec: $CO + 2H_2 \rightleftharpoons CH_2^* + H_2O$

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(2) $2A + D \rightarrow B + C$

$$r_A = \frac{-k P_D P_A^2}{(1 + K_A P_A + K_B P_B)^2}$$

$A + S \rightarrow AS$ Adsorption

$2AS + D \rightarrow BS + C + S$ Surface Reaction

$BS \rightleftharpoons B + S$ Desorpsion

Surface reaction is rate limiting,

$$r = k (C_{AS}^2 P_D)$$

For Adsorption & Desorpsion,

$C_{AS} = K_A P_A C_V$ $\quad C_{BS} = K_B P_B C_V$

Site Balance,

$$C_T = C_V + C_{AS} + C_{BS}$$

$$C_T = C_V + K_A P_A C_V + K_B P_B C_V$$

$$C_V = \frac{C_T}{1 + K_A P_A + K_B P_B}$$

$$r = k (C_{AS}^2 P_D)$$

$$= k (K_A^2 P_A^2 C_V^2 P_D)$$

$$\boxed{r = k \frac{K_A^2 P_A^2 P_D}{(1 + K_A P_A + K_B P_B)^2}}$$